Minimum Candidate Dimension for Mixed-Integer Optimization: Sharp Identification and Solver Certification

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Abstract

Repeated mixed-integer optimization often uses a ranked subset of entities to restrict each solve. The relevant quality object is not simply the size of a restricted model, but the minimum candidate dimension that attains a prescribed objective tolerance. For a nested candidate family we study $Q_{\mathcal{C}}(\theta;\tau) = \min\{B : J_\theta(\mathcal{C}_\theta(B)) - J_\theta^* \le \tau\}$. A support-entry characterization explains this spectrum, including ties among optimal solutions. Our principal result gives the sharp, possibly noncontiguous identified set for $Q_{\mathcal{C}}$ from globally valid but incomplete candidate and full-model primal–dual records, together with a necessary-and-sufficient point-identification criterion. Exactness is the specialization $\tau = 0$. Standard split-conformal calibration can control the required dimension marginally when its labels correspond to the mathematical target; a separate full-model bound is needed for instance-level solver certification. A full-prefix re-analysis of the same 80 held-out facility-location instances illustrates identification from final solver bounds while exposing numerical incompatibilities in some raw intervals. A conservative full-MIP-minus-LP lower-bound difference exceeds the 0.001 certification tolerance in all 80 distinct instances (240 scale–confidence records), explaining why that particular LP relaxation supplies no early certificate. Warm starts reduce full-solver search at moderate scales, but the audited certified candidate path remains slower than a cold full solve at $N = 20$–50. The paper thus separates structural candidate quality, statistical uncertainty, information-limited identification, and the cost of global certification.

Keywords: mixed-integer optimization; candidate dimension; partial identification; solver bounds; split conformal prediction; facility location.

1 Introduction

Repeated mixed-integer optimization (MIO) arises when related instances of a scheduling, location, or portfolio model are solved as their data change. Restricting available entities can simplify a solve, whether by physically removing variables or by fixing excluded blocks, but a restricted solve alone does not show that omitted entities were dispensable. A feasible candidate incumbent is an upper bound for a minimization problem, not evidence of its full-space optimality.

The missing object is the *minimum candidate dimension required for prescribed optimization quality*. Fix a nested ranking-based family of restricted feasible regions and write $J_\theta(\mathcal{C}_\theta(B))$ for the exact candidate optimum and J_θ^* for the full optimum. Their difference $g_\theta(B)$ is a nonincreasing quality–dimension frontier. Its generalized inverse $Q_{\mathcal{C}}(\theta;\tau)$ is the first budget whose quality loss is at most τ. In particular, the minimum exactness dimension is $Q_{\mathcal{C}}(\theta;0)$, not a second parallel target.

Even a complete description of this conceptual frontier does not mean it is observed. Solvers often provide only incumbent and global-bound records at a few budgets. A feasible incumbent

can be suboptimal within its candidate set; a full-model lower bound can be loose; and an unfinished solve may have no incumbent. Consequently, the earliest sufficient budget can remain ambiguous although all recorded bounds are valid.

We characterize that ambiguity sharply. Intersecting the candidate and full intervals with nesting yields monotone envelopes, but the attainable first-passage budgets are not generally a contiguous interval. We give a necessary-and-sufficient membership criterion, a constructive converse, and a two-sided condition for point identification. The theorem is sharp relative to the declared scalar-record information class; a method using full coefficients, cuts, or other structural information may deduce more.

Across instances, standard split conformal prediction can calibrate a budget for the random variable $Q_{\mathcal{C}}(\theta;\tau)$ when genuine target labels are available. This statistical layer is distinct from a deterministic, instance-level bound check. Operational labels based on agreement with a feasible reference are not silently reinterpreted as mathematical exactness labels. Likewise, choosing the full budget is a structural fallback, not by itself a numerical certificate.

The computational experiments make the separation concrete. An integer-data facility-location audit measures coverage against solver-audited lattice labels; a real-valued facility-location audit measures reference fidelity, certification, warm-start effects, and wall-clock time on locked instances. A portfolio MIQP is a second application. Candidate compression and reduced branch-and-bound work do not imply end-to-end certified acceleration when global lower-bound information is costly or weak.

Our contributions are fourfold:

1. We define the *Candidate Quality-Budget Spectrum* and characterize it by first entry of a τ-optimal entity support. This also exposes the complexity of finding an earlier alternative optimum even when one full optimum is known.
2. We give a sharp, potentially noncontiguous identified set from incomplete globally valid solver records and an exact point-identification criterion. This is the principal theoretical contribution.
3. We place conventional split-conformal budget calibration and deterministic solver certification in separate layers, with explicit reference-error, lower-bound, and candidate-incumbent qualifications.
4. We audit facility-location evidence for structural candidate restriction, LP-bound barriers, search reduction, and measured certified-path time, and retain portfolio MIQP as a distinct formulation rather than pooling incomparable outcomes.

We do not claim a new conformal principle, safe-screening rule, branch-and-bound method, universal candidate speedup, or investment superiority. The theoretical information result does not establish a fast general algorithm for computing every restricted optimum.

2 Related Work and Claim Boundary

Learned reduction is established: Triantafyllou and Papathanasiou [2024] predict dimension-reducing decisions in mixed-integer optimization, and Li et al. [2024] learn reduced MILP models. Predict-and-search methods use learned assignments or neighborhoods to find strong feasible solutions [Han et al., 2023, Cai et al., 2025]. Our target is different: for a *fixed nested entity family*, what is the minimum dimension meeting a specified objective threshold, and what can incomplete records identify about it? We make no priority claim for producing reduced models or candidate sets.

Deterministic safe screening excludes variables without losing globally optimal solutions. Examples include Fenchel-based screening for cardinality-constrained Poisson regression [Kurihara and Izunaga, 2025] and safe binary asset fixing for cardinality-constrained quadratic portfolio optimization using perspective relaxation, Fenchel duality, and valid bounds [Wada et al., 2026].

Our potentially risky ranked prefix is not such a safe rule. The quality of the prefix and the validity of a certificate remain separate questions.

Conformal optimization guarantees already exist: Ovalle et al. [2025] calibrate feasibility for learned MIO constraints, Clarke and Stellato [2025] calibrate performance of a learned hierarchical MIO method, including facility-location experiments, and Clarke and Stellato [2026] use conformal prediction with solver state to guide early stopping under a probabilistic quality guarantee. We instead calibrate the *required candidate dimension*; the sharp deterministic identified set is a different object from their feasibility, solution-quality, and stopping targets. Standard split-conformal rank theory supplies our statistical layer, not a new coverage principle.

The mathematical engine of interval intersection and monotonicity has substantial antecedents in partial identification [Manski and Tamer, 2002] and monotone function intervals [Yang and Zentefis, 2023]. Our claim is restricted to the particular MIO first-passage object, its noncontiguous scalar-record identified set, and its solver-bound point-ID condition. Likewise, alternative-solution and fixed-variable questions are familiar from backbone complexity [Kilby et al., 2005]; the supplied-optimum hardness proposition below is a boundary result, not a new complexity paradigm. These comparisons are based on a bounded primary-source audit, not an assertion that no equivalent theorem exists elsewhere.

3 Candidate Quality-Budget Spectrum

Fix an instance θ, a finite entity universe $\mathcal{N} = \{1, \dots, N\}$, a global variable block v_0, and entity blocks v_i with designated null values v_i^0. The full feasible set $\mathcal{F}_\theta$ is nonempty; its minimum $J_\theta^* = \min_{v \in \mathcal{F}_\theta} f_\theta(v)$ is finite and attained. Let $\mathcal{C}_\theta(B)$ be a fixed nested candidate family of cardinality B, with $\mathcal{C}_\theta(N) = \mathcal{N}$. An excluded entity is fixed to its null block, defining $\mathcal{F}_\theta(B) = \{v \in \mathcal{F}_\theta : v_i = v_i^0 \text{ for } i \notin \mathcal{C}_\theta(B)\}$. Assume a first feasible budget b_0 exists and each feasible restricted minimum is attained. Then $\mathcal{F}_\theta(B)$ is feasible for all $B \geq b_0$ and $J_B := \min_{v \in \mathcal{F}_\theta(B)} f_\theta(v)$ is nonincreasing, with $J_N = J_\theta^*$.

Define

$$g_\theta(B) = J_B - J_\theta^*, \qquad Q_{\mathcal{C}}(\theta; \tau) = \min\{B \in \{b_0, \dots, N\} : g_\theta(B) \leq \tau\}, \quad \tau \geq 0. \tag{1}$$

For a feasible v, let $S(v) = \{i : v_i \neq v_i^0\}$ and $\text{entry}_{\mathcal{C}}(S) = \min\{B \geq b_0 : S \subseteq \mathcal{C}_\theta(B)\}$. A mandatory set M_θ, when present, is contained in every candidate set; it does not by itself prove all reduced models feasible.

Theorem 1 (Quality–Support First-Entry Characterization)**.** *For* $\Omega_\theta^\tau = \{v \in \mathcal{F}_\theta : f_\theta(v) \leq J_\theta^* + \tau\}$ *and every* $\tau \geq 0$*,*

$$Q_{\mathcal{C}}(\theta; \tau) = \min_{v \in \Omega_\theta^\tau} \text{entry}_{\mathcal{C}}(S(v)). \tag{2}$$

Consequently $Q_{\mathcal{C}}(\theta; 0) = B_{\mathcal{C}}^\dagger(\theta)$*, the first exact prefix. The formula takes the minimum over* all *τ-optimal solutions and supports, not one selected incumbent or one selected full optimum.*

Proof. For each $B \geq b_0$, attained restricted optimality gives $g_\theta(B) \leq \tau$ if and only if some $v \in \mathcal{F}_\theta(B)$ has $f_\theta(v) \leq J_\theta^* + \tau$. By the null-block definition of $\mathcal{F}_\theta(B)$, this is equivalent to some $v \in \Omega_\theta^\tau$ satisfying $S(v) \subseteq \mathcal{C}_\theta(B)$. Minimizing the first admissible budget on both sides proves the identity; setting $\tau = 0$ proves the exactness specialization. □

For a top-ranked family with mandatory set M_θ, let $r_\theta(i)$ be the fixed rank among nonmandatory entities, and $\mathcal{C}_\theta(B) = M_\theta \cup \{i : r_\theta(i) \leq B - |M_\theta|\}$. The following formula also handles empty nonmandatory support via $\max \varnothing = 0$.

Corollary 2 (Ranked Support Depth and Pareto Steps)**.**

$$Q_{\mathcal{C}}(\theta;\tau) = \max\left\{b_0,\ |M_\theta| + \min_{v\in\Omega_\theta^\tau} \max_{i\in S(v)\setminus M_\theta} r_\theta(i)\right\}. \tag{3}$$

Moreover $Q_{\mathcal{C}}(\theta;\tau)$ is a nonincreasing, right-continuous finite step function of $\tau \geq 0$; its changes occur only at distinct candidate gaps $g_\theta(B)$.

Proof. The first prefix containing $S(v)$ is the displayed maximum. Substitution into Theorem 1 gives the first statement. The second follows from the finite family of weak inequalities $g_\theta(B) \leq \tau$ and $g_\theta(N) = 0$. □

Write $\Delta_{\mathcal{C}} = \min_{b_0\leq B<B_{\mathcal{C}}^\dagger} g_\theta(B)$, with $+\infty$ if $B_{\mathcal{C}}^\dagger = b_0$. When finite it equals $g_\theta(B_{\mathcal{C}}^\dagger - 1) > 0$, so $Q_{\mathcal{C}}(\theta;\tau) = B_{\mathcal{C}}^\dagger$ for $0 \leq \tau < \Delta_{\mathcal{C}}$. This is ordinary first-passage separation, not another theorem. The full $g_\theta(B)$ vector is conceptually well defined but is not recorded for every prefix in the locked experiments.

4 Sharp Identification from Incomplete Solver Records

Suppose only scalar, globally valid minimization bounds are observed: $L_B \leq J_B \leq U_B$ for $B = b_0, \ldots, N$, and $L_* \leq J_\theta^* \leq U_*$. A feasible, checked candidate incumbent supplies an upper bound; a candidate-specific global dual bound supplies a lower bound for that candidate, not automatically for the full model. An absent bound can be represented by an infinite endpoint. We assume the record family is jointly compatible and finite enough for the displayed constructions. The information class contains nesting but *not* the full model coefficients, shared cuts, or internal search trace.

Absorb the full bounds at N: $l_B = L_B, u_B = U_B$ for $B < N$, $l_N = \max(L_N, L_*)$, and $u_N = \min(U_N, U_*)$. Define

$$a_B = \max_{j\geq B} l_j, \qquad c_B = \min_{i\leq B} u_i, \qquad S_* = [a_N, c_N]. \tag{4}$$

Compatibility is equivalent to $a_B \leq c_B$ at every budget. The suffix maximum and prefix minimum are the sharp coordinatewise envelopes of a nonincreasing value profile. In particular, S_* is the sharp terminal-value interval, potentially tighter than $[L_*, U_*]$ alone.

Theorem 3 (Sharp Identification of the Candidate Quality Budget from Solver Records)**.** *Under the compatible scalar-record information class above, the exact identified set of $Q_{\mathcal{C}}(\theta;\tau)$ is*

$$\begin{aligned}\mathcal{Q}_\tau = {} & \{b_0 : S_* \cap [a_{b_0} - \tau, \infty) \neq \varnothing\} \\ & \cup \{r \in \{b_0+1, \ldots, N\} : S_* \cap [a_r - \tau, c_{r-1} - \tau) \neq \varnothing\}.\end{aligned} \tag{5}$$

Every listed budget is attained by a compatible monotone objective-value profile, and no unlisted budget is. The set can be noncontiguous; reporting only its endpoint interval is generally not sharp.

Proof. Let $j = (j_{b_0}, \ldots, j_N)$ be any nonincreasing profile compatible with the raw intervals and put $s = j_N$. For every B, later lower records imply $j_B \geq a_B$, while earlier upper records imply $j_B \leq c_B$; hence $s \in S_*$. If b_0 first passes, $j_{b_0} \leq s + \tau$ requires $s \geq a_{b_0} - \tau$. If $r > b_0$ first passes, then $a_r \leq j_r \leq s + \tau$ and $c_{r-1} \geq j_{r-1} > s + \tau$. Thus $s \in [a_r - \tau, c_{r-1} - \tau)$, proving necessity, including the strict predecessor boundary.

For sufficiency fix any s in the indicated intersection. For $B < N$ define the lower profile $m_B(s) = \max\{a_B, s\}$ and upper profile $M_B(s) = c_B$, and set $m_N(s) = M_N(s) = s$. Both are nonincreasing and satisfy every raw interval, since $a_B \leq c_B$, $s \leq c_N \leq c_B$, and the

endpoint bounds hold. For $r > b_0$, choose $j_B = M_B(s)$ when $B < r$ and $j_B = m_B(s)$ when $B \geq r$. The splice is nonincreasing because $c_{r-1} \geq \max\{a_r, s\}$. Every earlier budget fails since $c_B \geq c_{r-1} > s + \tau$; budget r passes since $a_r \leq s + \tau$. For $r = b_0$, use $m(s)$ throughout. Hence each and only each listed budget is attainable. Finally, any finite nonincreasing profile can be realized as the prefix-optimum sequence of a finite one-hot binary-choice MIO with option costs equal to its profile values. This last realization concerns the scalar information class, not a fixed application formulation. □

The sharp minimum and maximum possible gaps at $B < N$ are $\max\{0, a_B - c_N\}$ and $c_B - a_N$, respectively. They are achieved by the profiles in the proof. These extrema yield a particularly simple point-ID condition.

Corollary 4 (Necessary and Sufficient Point Identification)**.** *For $q \in \{b_0, \ldots, N\}$, $\mathcal{Q}_\tau = \{q\}$ if and only if*

$$(q = N \text{ or } c_q - a_N \leq \tau) \quad \text{and} \quad (q = b_0 \text{ or } a_{q-1} - c_N > \tau). \tag{6}$$

At $\tau = 0$, this identifies $B^\dagger_{\mathcal{C}}$. Exact optimization of every candidate prefix is not required if the recorded valid bounds already satisfy both tests.

Proof. The first inequality says every compatible profile passes at q; $q = N$ passes by definition. The second says every profile fails at the predecessor; it is vacuous for $q = b_0$. The displayed sharp gap extrema make both conditions necessary as well as sufficient. Monotonicity then propagates predecessor failure to all earlier budgets and passage at q to all later ones. □

Corollary 5 (Integer-Lattice Localization)**.** *Suppose the candidate* optimal values *J_B all belong to a proven common lattice $\alpha_0 + h\mathbb{Z}$, $h > 0$. After establishing mathematically valid objective intervals, map each finite lower endpoint to $\alpha_0 + h\lceil (l_B - \alpha_0)/h \rceil$ and each finite upper endpoint to $\alpha_0 + h\lfloor (u_B - \alpha_0)/h \rfloor$, then recompute the envelopes. Theorem 3 and Corollary 4 apply exactly to lattice-valued profiles, with τ equivalent to $h\lfloor \tau/h \rfloor$. Raw floating-point solver displays are not, without numerical validation, exact lattice endpoints.*

Proof. Lattice intersection preserves all compatible true optimal values and excludes all others. Suffix maxima, prefix minima, and the upper-before/lower-after splice remain on the common lattice. Differences of lattice values are multiples of h, giving the tolerance reduction. □

For a transparent noncontiguous example, budgets $1, 2, 3$, incumbent values $(2, 3, 2)$ with a common error cap 1, and a full lower bound 1 admit compatible true profiles $(2, 2, 2)$ and $(2, 2, 1)$. Their exactness budgets are 1 and 3; budget 2 is impossible under the same records. Appendix B gives the interval and construction details. This is a deterministic identified set, not a statistical confidence set.

5 Proxy Translation, Calibration, and Certification

The target $Q_{\mathcal{C}}(\theta; \tau)$ is mathematical. Practical reference values, finite candidate incumbents, and full lower bounds reveal different first-passage predicates. The following identities make their relationship explicit.

Proposition 6 (Reference and Certification Translation)**.** *Let a feasible full-universe reference have value $J_{\text{ref}} = J^*_\theta + \delta$, $\delta \geq 0$. Its ideal reference-fidelity budget at tolerance ε_{ref} is*

$$R^{\text{or}}_{\text{ref}} = \min\{B : J_B - J_{\text{ref}} \leq \varepsilon_{\text{ref}}\} = Q_{\mathcal{C}}(\theta; \varepsilon_{\text{ref}} + \delta). \tag{7}$$

For a globally valid full lower bound $L = J^*_\theta - \eta$, $\eta \geq 0$, define the ideal exact-candidate certificate budget $R^{\text{or}}_{\text{cert}} = \min\{B : J_B - L \leq \varepsilon_{\text{cert}}\}$, with empty minimum $+\infty$. Then

$$R^{\text{or}}_{\text{cert}} = Q_{\mathcal{C}}(\theta; \varepsilon_{\text{cert}} - \eta), \tag{8}$$

where $Q_{\mathcal{C}}(\theta; t) = +\infty$ for $t < 0$. In particular, if $\eta > \varepsilon_{\text{cert}}$, this fixed bound cannot certify even a true full optimum at the requested tolerance.

Proof. The two residuals are $J_B - J_{\text{ref}} = g_\theta(B) - \delta$ and $J_B - L = g_\theta(B) + \eta$; rearrange their weak inequalities. At $B = N$ the second residual is η. □

If a feasible candidate incumbent is $U_B = J_B + \zeta_B$ with $\zeta_B \geq 0$, then

$$U_B - L = g_\theta(B) + \eta + \zeta_B. \tag{9}$$

An observed incumbent pass is no earlier than its exact-candidate pass. If a uniform finite error $\zeta_B \leq \bar{\zeta}$ holds at every budget, the first observed pass lies between $Q_{\mathcal{C}}(\theta; t)$ and $Q_{\mathcal{C}}(\theta; t - \bar{\zeta})$ for the corresponding shifted threshold t; missing incumbents invalidate that finite-error upper bound. The exact sharp treatment of heterogeneous per-budget information is Theorem 3, not this elementary sandwich.

5.1 Split-Conformal Calibration of Required Dimension

Assume a ranking and any predictive score model are fixed independently of calibration examples. Let $R_j = Q_{\mathcal{C}}(\theta_j; \tau)$ be genuine mathematical labels on n exchangeable calibration instances, with an exchangeable future instance. For a marginal budget rule, let $k = \lceil (n+1)(1-\alpha) \rceil$. If $k > n$, set the budget to N; otherwise let $\widehat{B}_\alpha$ be the kth smallest calibration label (clipped to the valid budget range). Standard conservative split-conformal rank reasoning [Lei et al., 2018] gives

$$\Pr\{Q_{\mathcal{C}}(\theta_{\text{new}}; \tau) \leq \widehat{B}_\alpha\} \geq 1 - \alpha. \tag{10}$$

Ties do not weaken this lower bound. The probability is marginal over the exchangeable new instance and calibration data, not conditional on its features. An independently fitted predictor $\widehat{q}(X)$ may instead use residuals $R_j - \widehat{q}(X_j)$ and the same rank rule; ceiling and clipping of the real threshold are conservative. This is an application of established conformal prediction, not a new conformal theorem.

There are three distinct label regimes. Genuine $Q(0)$ labels, or labels known to upper-bound them, support a marginal *exactness-dimension* statement. Genuine $Q(\tau)$ labels support marginal τ-quality of an *exactly solved* restricted model. Operational reference-fidelity labels calibrate only that operational target unless a separate conservativeness argument links them to $Q(0)$ or $Q(\tau)$. If $R \geq Q_{\mathcal{C}}(\theta; 0)$ with probability at least $1 - \beta$ and the conformal rule covers R with probability at least $1 - \alpha$, the union bound yields exactness-budget coverage at least $1 - \alpha - \beta$; no independence is required. Estimating this β post hoc from the test records would not establish such a guarantee. A numerical solver certificate remains a separate check in every regime.

5.2 Bound-Driven Decision and Recovery

At a fixed budget, a checked feasible incumbent U_B and a globally valid full lower bound L_{full} establish instance-level absolute objective quality only if $U_B - L_{\text{full}} \leq \varepsilon_{\text{cert}}$. A candidate-restricted lower bound is not a full lower bound. Using valid candidate and full records, Corollary 4 can identify a *minimum dimension*; an ordinary incumbent–full-bound check certifies the quality of one returned solution. These are related but distinct claims. PASS(B) may be inferred from $c_B - a_N \leq \tau$, FAIL(B) from $a_B - c_N > \tau$ for $B < N$, and otherwise the scalar records are unresolved. An unresolved status must not be called a pass.

If a restricted solve is unresolved or too low-quality, a geometric schedule with $\rho > 1$,

$$B_{j+1} = \min\{N, \max(B_j + 1, \lceil \rho B_j \rceil)\},$$

eventually reaches N. Before the cap its number of expansions to any target budget is logarithmic in the budget ratio, and the cumulative allowed entity counts are geometric-series bounded.

These are *structural accounting* statements, not counts of physically instantiated variables or bounds on branch-and-bound runtime or certification cost. At N, a solver still needs an adequate valid primal–dual gap; a time limit or the full candidate set alone does not certify a numerical objective. The full solve can be warm-started by a candidate incumbent only after full-model feasibility is checked.

6 Computational Boundary of Candidate Quality Budgets

Supplying the full optimum does not automatically locate the first optimal prefix: a different optimum may enter earlier than the supplied one. The following narrowly scoped result makes that distinction precise.

Proposition 7 (Supplied-Optimum Restricted-Prefix Hardness)**.** *For bounded-coefficient 0–1 linear minimization models, deciding whether $Q_{\mathcal{C}}(\theta; 0) \leq k$ is NP-complete even when a fixed ranking, $J^*_\theta = 0$, and a globally optimal vector are supplied, every prefix is feasible, and $N - k = 1$. The analogous decision at the fixed tolerance $\tau = 1$ is NP-complete on the same construction.*

Proof. Appendix C gives the full polynomial reduction from 3SAT and membership witness. Its hard step is restricted-prefix alternative-optimum feasibility: fixing the one last-ranked entity to zero exposes the SAT constraints while a supplied zero-cost full optimum remains available. This is not a hardness result for searching an already known numerical prefix-value sequence. □

Given exact prefix values and J^*_θ, one scan recovers the full spectrum, and an exact monotone predicate permits classical binary search for one $Q_{\mathcal{C}}(\theta; \tau)$. LP, min-cost-flow, and standard matroid formulations with polynomially solvable restricted prefixes therefore yield polynomial computation by solving the required prefixes. This boundary does not imply that general MIO prefix values are easy. The set-cover CFLP specialization is relegated to Appendix C.

7 Two Application Families

Capacitated facility location. A binary opening decision is the entity block for each of N facilities; continuous flows connect retained facilities to M customers. The full and prefix models minimize opening plus transportation cost subject to demand and capacity. Both audited protocols use a complete bipartite service network and nonnegative flows, with no compatibility arcs or minimum shipment rules. A prefix is feasible exactly when its total capacity reaches total demand, so b_0 is the first ranked prefix satisfying that inequality. The real-valued scaling protocol uses a fixed, unfitted efficiency ranking $f_i/s_i + M^{-1} \sum_j c_{ij}$; the integer-data protocol has a separate exactness-label purpose. These are different experimental targets, not interchangeable oracle implementations.

Multi-period portfolio MIQP. Each risky asset is an entity block containing holdings, trades, absolute-trade auxiliaries, and binary holding/trading indicators across the planning horizon. Cash and aggregate accounting variables remain global blocks. The historical holdings exceeding 10^{-6} are included in every candidate set, and the remaining risky assets enter by the fixed ranking. This mandatory-state construction avoids dropping held positions, but it does not by itself establish feasibility of every restricted portfolio MIQP. The model retains quadratic risk, expected returns, fixed and proportional transaction costs, cardinality, position bounds, liquidity, and turnover constraints. A valid full-model relaxation or full MIP bound is needed for certification. Appendix F records the detailed formulation and accounting. Perspective strengthening is used only as a standard means of tightening a relaxation, not as a claimed new contribution.

8 Locked Computational Evidence

The experiments ask whether candidate dimension is observable, whether a lower bound can certify it or a solution obtained at it, and whether reduced search pays for its own certification. The original statistical outcomes are from locked artifacts; the separately labeled full-prefix re-analysis runs new solves only on the same 80 locked test instances to reveal trajectories absent from those artifacts. No new statistical instance or tuning was introduced. The two CFLP protocols and portfolio formulation have different outcome semantics.

8.1 Protocol and Claim Semantics

Mathematical $Q_{\mathcal{C}}(\theta; 0)$ uses exact candidate and full objective values. *Reference fidelity* compares a candidate objective or incumbent with a saved feasible full reference at a stated tolerance; unless the reference error is controlled through Proposition 6, this is not mathematical equality. *Certified* $\varepsilon_{\text{cert}}$*-quality* uses a feasible incumbent and a globally valid full-model lower bound satisfying an absolute gap. A solver's `OPTIMAL` status and displayed floating-point bounds are numerical evidence under its tolerance and model semantics, not rational-arithmetic proof. The reported path wall-clock components include ranking, restricted solves, relaxations, recovery, and full certification where those stages are charged; Section 8.6 discloses the separately executed validation overhead.

The real-valued CFLP experiment has $(N, M) = (8, 15), (20, 20), (30, 25), (50, 30)$, 32 calibration and 20 held-out instances per scale, and $\alpha \in \{0.20, 0.10, 0.05\}$. Its label is one-sided reference fidelity at $\varepsilon_{\text{ref}} = 10^{-4}$; the certification tolerance is $\varepsilon_{\text{cert}} = 10^{-3}$. The same 80 distinct held-out generation seeds occur at all three α values, yielding 240 scale–confidence–test records, not 240 independent problem draws. The integer-data CFLP audit separately uses $(N, M) = (10, 15)$ and 30 replications with 20 calibration and 20 test instances each. Portfolio results use their own solver and temporal protocols and are not pooled with CFLP timings.

8.2 Integer-CFLP Solver-Audited Lattice Labels

Integer opening costs, unit transportation costs, demands, and capacities imply that for each fixed binary opening vector the transportation LP has an integral optimal extreme point. Thus each *optimal prefix value* is integer, although arbitrary feasible continuous flows may have fractional cost. Distinct mathematical optimal values are separated by at least $h = 1$, giving the lattice basis for $Q(0)$ labels. This mathematical fact is separate from floating-point numerical certification.

The locked label code accepts a feasible solver result if the reported absolute `UB-LB` is less than 0.95, then applies `round(UB)`; every saved full record has zero displayed gap and an integer displayed incumbent. It did *not* save independently outward-rounded candidate/full intervals or a near-integer residual for every prefix. Hence we call these *integer-lattice solver-audited labels under the stated numerical criterion*, not exact-arithmetic certificates. Every one of the 1,200 saved calibration/test instance labels has `full_exact_certified=True` and `label_status=OK` under that code path. This disclosure is essential to interpreting the following coverage outcomes.

Table 1: Locked integer-CFLP solver-audited exactness-label experiment: 30 replications, 20 calibration and 20 test cases per replication. Bootstrap intervals summarize the mean across replications; they are not conformal coverage guarantees.

α	Nominal target	Mean test coverage	95% bootstrap CI of mean	Mean selected B/N
0.20	80%	84.83%	[80.00%,89.33%]	88.67%
0.10	90%	96.50%	[94.00%,98.67%]	97.00%
0.05	95%	98.83%	[96.83%,100.00%]	99.00%

These finite-test means are consistent with marginal split-conformal coverage under exchangeability and the label semantics just described. The mathematical guarantee does not require each individual replication to meet its nominal target. The available artifact records J^* and the first matching prefix budget, not a complete J_B trajectory; it therefore cannot support a full empirical $Q(\tau)$ staircase or $\Delta_{\mathcal{C}}$ distribution.

8.3 Real-Valued CFLP Reference-Fidelity Compression

In the real-valued protocol, the calibrated initial budget targets an operational reference-fidelity label, *not* $Q_{\mathcal{C}}(\theta; 0)$. At $\alpha = 0.20$, initial dimension ratios B_1/N are $0.875, 0.750, 0.733, 0.740$ for $N = 8, 20, 30, 50$, respectively. The initially restricted cases in these four cells all recover to the full universe under the locked operational policy. At higher confidence, the selected budget grows and can start at N, particularly for $N = 8$. These statements describe structural selection and recovery, not numerical certification. The complete pre-specified grid is in Appendix E.

8.4 The LP Certification Barrier

Proposition 6 predicts a bound-quality obstruction. Let $L_{\text{full}}^{\text{MIP}}$ be the saved globally reported full-MIP lower bound and $L_{\text{full}}^{\text{LP}}$ the saved lower bound from the particular full-universe LP relaxation. Conditional on their numerical validity for the same real-valued objective,

$$\underline{\eta}_{\text{LP}} := L_{\text{full}}^{\text{MIP}} - L_{\text{full}}^{\text{LP}} \leq J_\theta^* - L_{\text{full}}^{\text{LP}} =: \eta_{\text{LP}}. \tag{11}$$

The full reference MIP and full LP both report `OPTIMAL` in all 240 Phase B records. A read-only aggregation gives $\underline{\eta}_{\text{LP}} > 0.001 = \varepsilon_{\text{cert}}$ in *80/80 distinct held-out instances*, represented three times each across confidence levels (*240/240* scale–confidence records); none has $\underline{\eta}_{\text{LP}} \leq 0.001$. By scale the minimum difference is 51.7479 ($N = 8$), 19.5932 ($N = 20$), 18.2590 ($N = 30$), and 12.3130 ($N = 50$), with 20 distinct instances per scale. Therefore this *fixed LP relaxation* cannot certify even a true optimum at the requested absolute tolerance in the saved cases, assuming valid reported bounds. The saved full incumbent–bound difference is at most 4.55×10^{-13}, but the argument uses the lower bound and does not presume an independent exact real-valued oracle. Stronger relaxations or cuts could change the result.

All 1,440 LP-bearing B.5 repetition rows also have saved `full_lb-lp_lb`> 0.001; these are repeated calls on the locked sample, not additional independent instances. Zero early LP certificates is therefore explained by the fixed-bound barrier for the whole audited Phase B sample, rather than inferred merely from aggregate runtime. The LP solve still incurs cost before full-MIP fallback in paths that attempt it.

8.5 Locked-Instance Full-Prefix Re-analysis of Quality-Budget Identification

The original Phase B records did not save complete prefix trajectories. We therefore reran the *same 80 distinct held-out real-valued CFLP instances*, reconstructed from their locked seeds, generator, efficiency ranking, and candidate family, at every feasible prefix. These are new solver runs to expose missing trajectories, *not* a new statistical sample or a tuned extension of the original conformal test. The inherited Phase B solver uses Gurobi 13.0.3, automatic threads, relative `MIPGap=1e-5`, a 10-second limit at proper candidate prefixes, and a 15-second full-prefix limit. It creates the full pre-presolve variable array and fixes excluded blocks to zero. The 1,414 new prefix records store candidate incumbents, global solver bounds, statuses, times, and the original and rerun full-model records in a separate hashed artifact. We did not rerun the original LP-barrier experiment.

We prespecified $\tau \in \{0, 10^{-4}, 10^{-3}\}$ before examining the new outputs. For each prefix, the reported bounds imply a candidate-gap interval $[\max\{0, L_B - U_N\}, U_B - L_N]$ when numerically valid; the full-prefix gap is identically zero. Applying Equation (5) to final prefix-bound records

yields a singleton identified set in 57 of 80 instances at each positive prespecified tolerance. In 51 of those 57, it first becomes a singleton after fewer than all proper candidate prefixes have been solved, with the terminal full record supplied at the outset. This is *prefix-bound identification*, not early-time solver identification or a measured acceleration. All new prefix solves report `OPTIMAL`, but 23 instances have incompatible raw floating-point intervals under the exact monotonicity test $a_B \leq c_B$; 21 conflicts are at most 1.6×10^{-12} and two reach approximately 4.2×10^{-7} and 5.5×10^{-6}. We neither project nor silently repair those intervals, and make no point-identification claim for those 23 instances.

At $\tau = 0$, literal reported endpoints yield 49 singleton, eight multiple, and 23 incompatible sets. Three of the eight multiple sets are noncontiguous, but their distinctions depend on solver-bound differences of floating-point scale, so they are not robust empirical demonstrations of mathematical noncontiguity. The constructed $\{1, 3\}$ example remains the sharpness witness. These diagnostics condition on the global validity and common-objective interpretation of solver bounds; they are not pooled with the original coverage estimates.

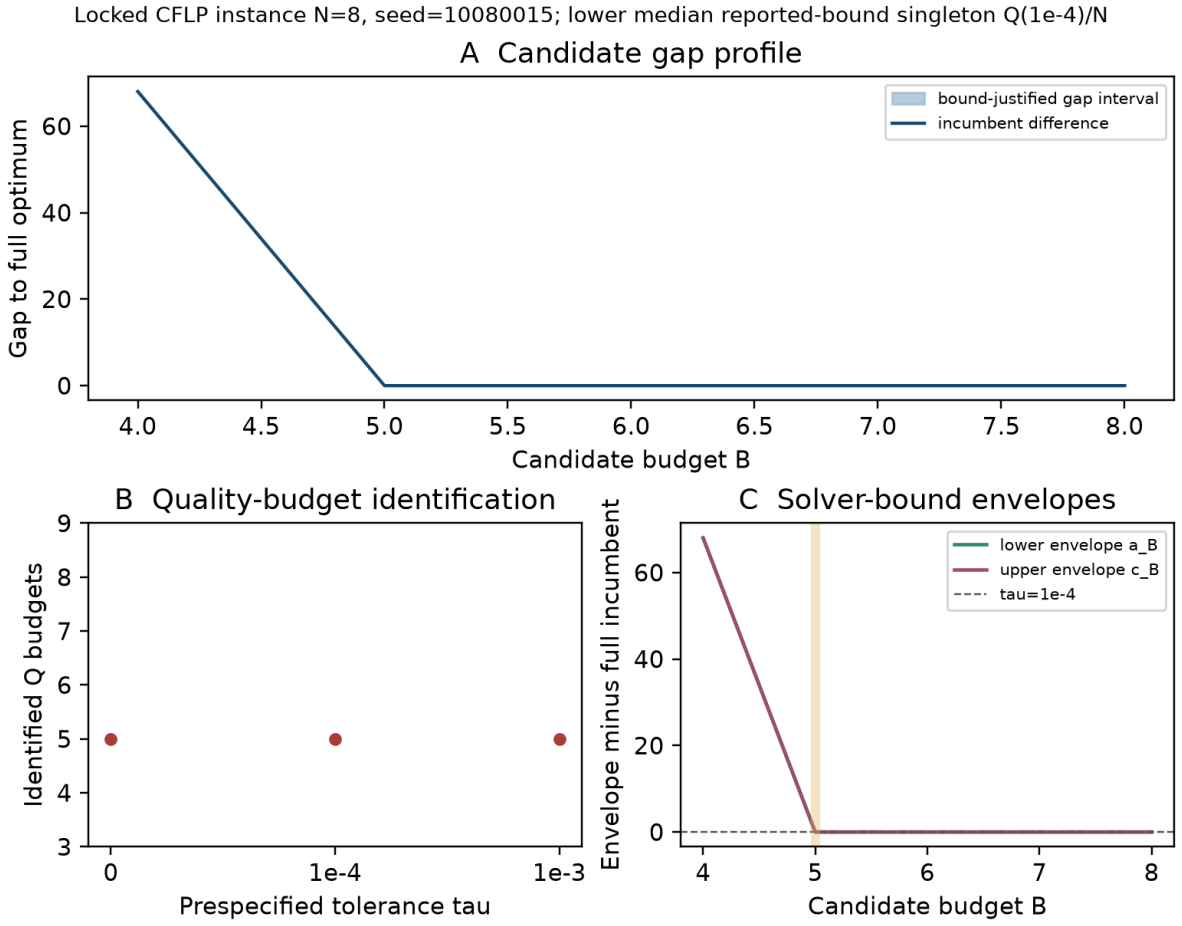


Figure 1: Prespecified theory–evidence display for locked instance seed 10080015. The selection is the lower median $Q(10^{-4})/N$ among the 57 instances with a singleton set under literal reported bounds; 23 numerically incompatible instances cannot enter that selection. Panel A shows incumbent differences and bound-derived gap intervals, B shows identified budgets at the three prespecified tolerances, and C shows candidate lower/upper envelopes with the 10^{-4} identified budget highlighted. This is a final-prefix-bound diagnostic, not a dynamic solver trace.

8.6 Warm Starts and Operational Certification

The final four-path B.5 audit uses the same CFLP formulation and locked held-out instances, Gurobi 13.0.3, one thread, relative `MIPGap=0`, absolute `MIPGapAbs=0.001`, a 15-second full-MIP limit, and a 10-second candidate-MIP limit. Path A is a cold full MIP. B is candidate–LP–cold full fallback; C is candidate–LP–warm full fallback; D is candidate–direct warm full fallback. The candidate incumbent is expanded to the full variable vector and checked against full-model feasibility before use as a start. In this implementation candidate restriction fixes excluded facility and shipment variables to zero in an otherwise full-size pre-presolve model; physical variable removal is not established. Each timed solve includes model construction and optimization, but explicit MIP-start validation occurs outside the timed components. The reported path totals are logical sums of selected components from interleaved experimental calls, not perfectly inclusive contiguous wall-clock pipeline measurements. Each path is invoked three times; an instance-level median is the timing observation, and paired bootstrap intervals resample 20 instances, not 60 calls. The reported speedup is the ratio of mean A time to mean candidate-path time.

Table 2: Locked $\alpha = 0.20$ certified CFLP path runtime, seconds (mean of 20 instance-level three-run medians). Explicit MIP-start validation is excluded from the timed components. D is the fastest candidate path but remains slower than A.

N	A: cold full	B: LP+cold	C: LP+warm	D: direct warm	A/D	95% paired CI
8	.0163	.0366	.0360	.0327	.499	[.414,.602]
20	.0517	.0981	.0937	.0879	.588	[.521,.669]
30	.0881	.1664	.1584	.1519	.580	[.525,.638]
50	.1359	.2711	.2525	.2378	.572	[.524,.630]

Of 1,200 actual warm fallbacks, 1,182 (98.5%) generated a new incumbent in the solver log. At $\alpha = 0.20$, mean full-solve nodes fell from 22.75 to 19.90 ($N = 20$), 43.05 to 34.85 ($N = 30$), and 90.75 to 73.45 ($N = 50$), comparing cold A with direct-warm D. These are search reductions, not net time gains: the candidate solve and, in B/C, the LP attempt exceed the saved full-certification time. Every final B.5 path passed its solver-reported absolute gap check; the zero LP early-certificate count means full-MIP bound work supplied the operational certificate. When calibration starts at N (the $N = 8$ high-confidence cells), a candidate solve is already full-dimensional and a near-unit D/A ratio is not a warm-start benefit. Reference-assisted candidate-only timings that omit the paid full reference solve are diagnostic and are not mixed with these certified totals.

8.7 Portfolio MIQP Provenance Boundary

The multi-period portfolio MIQP supplies a second application formulation, not a controlled common-instance solver benchmark or a financial-performance hypothesis. Archived strategy-level aggregate timing tables have three conflicting lineages without retained per-epoch solver telemetry or an immutable run–configuration link. We therefore omit their numerical timing and speed comparisons. Moreover, each backtest strategy evolves its own holdings and wealth, so the archived cross-strategy objective-match field is not a within-instance exact-recovery rate and is not used as a mathematical $Q_{\mathcal{C}}$ label. Chronological market dependence does not establish the exchangeability premise of Section 5.1. Appendix F records the formulation and the standard relaxation comparison; neither portfolio evidence nor its legacy aggregate columns enter the CFLP identification or coverage estimates.

9 Discussion

Four difficulties should be kept distinct. *Structural candidate quality* is encoded by $g_\theta(B)$ and $Q_{\mathcal{C}}(\theta;\tau)$ for a fixed family; rankings can change that frontier but the present paper does not optimize rankings. *Statistical uncertainty* concerns how a dimension calibrated on past instances behaves on a new exchangeable instance; conformal coverage is marginal and depends on label semantics. *Incomplete solver information* means that even for one fixed instance the required dimension may only be set-identified, perhaps with holes; valid heterogeneous primal–dual records can sometimes point-identify it without exact solves at every prefix. *Global certification cost* is the expense of proving that a returned incumbent is close enough to the full optimum. The LP-barrier audit shows a setting in which the chosen relaxation is mathematically too loose for the requested tolerance, however accurately the candidate was selected.

The headline theorem is intentionally an information result, not a claimed breakthrough in interval mathematics. Its sharpness assumes only scalar records and monotonicity, so model-specific constraints can strengthen identification. A three-way bound-driven implementation may remain unresolved under time limits and has no optimal query policy or general finite-time guarantee. The original locked artifacts do not contain historical partial candidate bounds or complete prefix trajectories. The separately labeled T6.1 re-analysis now exposes final prefix records on the same instances, but it does not recover intermediate-time traces or measure early-time point identification. Its raw numerical incompatibilities also prevent treating every reported solver endpoint as an exact mathematical interval. No dynamic speed advantage or exact noncontiguity frequency is inferred from the new runs.

Numerically, solver-reported bounds inherit feasibility, integrality, and optimality tolerances. The integer-CFLP objective lattice is exact mathematics for *optimal* transportation subproblems; the archived prefix labels are solver-audited under a 0.95 reported-gap rule, without per-prefix outward-rounded records. The real-valued LP conclusion is conditional on global-bound validity. Computationally, candidate solve time, relaxation time, fallback time, and start validation all belong in an operational comparison; the saved B.5 totals omit the separately executed start-validation overhead. This omission cannot reverse the reported negative candidate-versus-cold result but prevents calling those totals fully inclusive end-to-end measurements. The negative certified-speed result is confined to these locked scales and solvers; it does not imply that other relaxations or larger problems cannot benefit. A portfolio ranking learned under distribution shift would also need fresh coverage diagnostics or a different statistical guarantee.

10 Conclusion

Minimum candidate dimension is a meaningful optimization-quality object: it records how far a nested restriction must expand to meet a tolerance, with exactness as $\tau = 0$. From incomplete but globally valid solver information, that dimension can be only partially identified; its sharp identified set need not be contiguous. Candidate and full primal–dual records can sometimes identify it exactly without solving every prefix. Across instances, established split-conformal calibration can control the required dimension marginally when labels represent the intended target. Instance-level certification additionally requires a sufficiently strong full-universe lower bound. In the locked facility-location audit, the particular LP relaxation cannot meet the requested certificate tolerance in any of 80 distinct held-out instances (240 scale–confidence records), and candidate restriction plus warm-start search reductions do not become certified wall-clock acceleration at $N = 20$–50 in the measured path components. The relevant end product is therefore a transparent separation of quality, information, coverage, and certification cost, not an unconditional speedup claim.

A Additional Proofs and Structural Consequences

Separation and proxy equality. If $B^\dagger_{\mathcal{C}} > b_0$, every earlier finite prefix has strictly positive gap by definition and attained optimality. Since $g_\theta(B)$ is nonincreasing, the smallest pre-exact gap is $g_\theta(B^\dagger_{\mathcal{C}} - 1) > 0$. For any $0 \le \tau < \Delta_{\mathcal{C}}$, no earlier prefix passes and $B^\dagger_{\mathcal{C}}$ does, proving $Q_{\mathcal{C}}(\theta;\tau) = B^\dagger_{\mathcal{C}}$. At $\tau = \Delta_{\mathcal{C}}$ the immediate predecessor passes, so the inequality is sharp. If $B^\dagger_{\mathcal{C}} = b_0$, the result is vacuous and $Q_{\mathcal{C}}(\theta;\tau) = b_0$ for all nonnegative tolerances. Substituting $\tau = \varepsilon_{\mathrm{ref}} + \delta$ in this statement specifies when an oracle reference-fidelity label equals exactness; no mere feasibility of the reference establishes the required separation event.

Finite incumbent first-pass bounds. Fix a comparator and write its oracle threshold as t. If $U_B = J_B + \zeta_B$ is a feasible candidate incumbent with $\zeta_B \ge 0$, an observed pass requires $g_\theta(B) + \zeta_B \le t$; hence its first pass cannot precede $Q_{\mathcal{C}}(\theta;t)$. If all candidate solves have a finite common bound $\zeta_B \le \bar{\zeta}$, every budget with $g_\theta(B) \le t - \bar{\zeta}$ necessarily passes, so the first observed pass is at most $Q_{\mathcal{C}}(\theta; t - \bar{\zeta})$. For $t - \bar{\zeta} < 0$ the upper bound is $+\infty$ and says nothing. A missing feasible incumbent must not be assigned a finite ζ_B. For a lower-bound comparator, the residual decomposition is Equation (9); for a feasible reference, replace L by J_{ref} and shift the threshold accordingly.

Coverage transfer and integer rounding. Let R be an exchangeable operational label and let $\widehat{B}$ cover it with probability at least $1 - \alpha$. If $\Pr\{R \ge Q_{\mathcal{C}}(\theta;0)\} \ge 1 - \beta$, the intersection of this event with $\{R \le \widehat{B}\}$ has probability at least $1 - \alpha - \beta$, and on the intersection $Q_{\mathcal{C}}(\theta;0) \le \widehat{B}$. This is an elementary union bound; it does not make proxy labels automatically conservative. For an adaptive residual threshold $T = \widehat{q}(X) + q_\alpha$, the event $Q_{\mathcal{C}}(\theta;\tau) \le T$ implies $Q_{\mathcal{C}}(\theta;\tau) \le \lceil T \rceil$, but the reverse implication can fail. Conservative ceiling and clipping therefore preserve the rank lower bound; equality of the two events must not be asserted.

Geometric structural accounting. For $B_{j+1} = \min\{N, \max(B_j + 1, \lceil \rho B_j \rceil)\}$, $\rho > 1$, uncapped budgets grow at least geometrically, so the number of uncapped expansions needed to reach a target $q > B_1$ is at most a ceiling of $\log_\rho(q/B_1)$, up to the final rounding/cap convention. If B_t is the last uncapped budget, the reverse geometric series bounds $\sum_{j \le t} B_j$ by a constant multiple of B_t depending on ρ. This counts allowed entity blocks, not necessarily physically created solver variables; no branch-and-bound node or time bound follows.

B Sharpness, Boundary Cases, and Information Scope

For the three-budget example in Section 4, the raw incumbent/error-cap intervals are $[1,2]$, $[2,3]$, and $[1,2]$, with full lower bound 1. After intersection at N, the suffix lower envelope is $(2,2,1)$, the prefix upper envelope is $(2,2,2)$, and $S_* = [1,2]$. At $\tau = 0$, Equation (5) admits 1 by $s = 2$ and 3 by $s = 1$. For $r = 2$ it requires simultaneously $s \ge a_2 = 2$ and $s < c_1 = 2$, which is impossible. Thus $\mathcal{Q}_0 = \{1,3\}$, not $[1,3] \cap \mathbb{Z}$. The compatible objective profiles $(2,2,2)$ and $(2,2,1)$ realize the two endpoints with the same scalar reports.

To realize a general finite nonincreasing compatible profile $j_{b_0} \ge \cdots \ge j_N$ as an MIO, introduce a one-hot binary option for each budget and assign option B cost j_B; the candidate prefix permits precisely the options with index at most its budget. Its minimum is j_B. Mandatory null entities can pad the family to a desired feasibility floor. This construction proves scalar-record observational sharpness across broad finite MIOs; it does not preserve the detailed coefficients of any particular CFLP, portfolio model, or solver trace. Incompatible raw bounds, for which some $a_B > c_B$, signal erroneous model identity, invalid numerical bounds, or a recording failure rather than an empty scientific conclusion.

At $\tau = 0$ the strict predecessor failure in Equation (6) cannot be changed to a weak inequality: equality permits an exact predecessor. When the candidate budget is N, the full candidate passes by definition even if full bounds are wide; numerical solution certification remains unresolved until a sufficient absolute primal–dual gap is established. These are different uses of the word "pass."

C Complexity Reductions and Tractable Cases

Full proof of Proposition 7. Given a 3-CNF formula ϕ in binary variables $x_1, \ldots, x_n$, introduce binary y, p, z. For each clause C, let $T_C(x)$ be the sum of its three literal truth indicators, using x_i for a positive literal and $1 - x_i$ for a negative literal. Construct

$$\min 2y \quad \text{subject to} \quad y + z + T_C(x) \geq 1 \ (C \in \phi), \qquad p \geq 1 - y - z, \quad p \leq 1 - y, \quad p \leq 1 - z. \tag{12}$$

Use entity order $(y, x_1, \ldots, x_n, p, z)$, $N = n + 3$, and test $k = N - 1 = n + 2$. A vector with $y = 1$ and every other variable zero is feasible in every prefix, so all prefixes are feasible. A full vector with $z = 1$ and all other variables zero has objective zero; because the objective is nonnegative, it is a supplied, provably global optimum and $J^* = 0$.

At any prefix excluding z, a zero-objective vector must have $y = z = 0$. The marker inequalities then force $p = 1$; therefore a zero-cost solution cannot exist before p enters at budget k. With $y = z = 0$, every clause inequality becomes $T_C(x) \geq 1$, exactly a satisfying assignment of ϕ. If ϕ is satisfiable, the first zero-cost prefix is k; otherwise it is N. Since every nonzero feasible objective is 2, the same dichotomy holds at the fixed quality tolerance $\tau = 1$. The reduction is polynomial, uses bounded-magnitude coefficients, and excludes only one entity at the tested prefix. A feasible prefix-k binary vector of value at most the supplied threshold is a polynomial NP witness, proving NP-completeness for the restricted class. It is the alternative-optimum restricted-prefix feasibility query, not difficulty of locating an index after exact values are known.

Facility-location specialization (supporting only). A nonmetric unit-demand CFLP can encode SET COVER at an input-dependent tolerance K: use a fallback facility with opening cost $K + 1$, a facility of opening cost 1 per set, and a last-ranked universal facility of cost zero; noncovered assignment costs $K + 1$ and all capacities allow all clients. The universal facility supplies a zero-cost full optimum, the fallback makes all prefixes feasible, and a prefix solution of cost at most K exists exactly when at most K set facilities cover all clients. This does not establish a fixed-tolerance CFLP theorem, metric-CFLP hardness, or a logarithmic approximation bound for the rank index.

Polynomial boundary. If exact optimization at every candidate prefix is polynomial (for example an LP, a network-flow restriction with integral data, or a matroid optimization family that retains its tractable form under the prefix), at most $N - b_0 + 1$ ordinary solves produce the entire finite spectrum. Given J^* and only a Boolean exact predicate $g(B) \leq \tau$, binary search uses logarithmically many prefix queries. With p binary variables and a polynomial LP after fixing them, exhaustive 2^p enumeration gives a familiar fixed-binary-dimension upper bound. None of these are presented as a new search algorithm.

D Solver Records, Algorithmic Use, and Numerical Validity

All interval claims require *global* bounds for the same exact objective, units, and candidate/full feasible sets. A local branch-and-bound node bound, unchecked infeasible start, relative gap alone, or solver status text alone is insufficient. An implementation can absorb each newly valid candidate and full bound, compute Equation (4), retain the full set in Equation (5), and return

a budget only when Corollary 4 holds. Otherwise it should report the current identified set and UNRESOLVED. Valid bound tightening can only shrink that set. A timeout may leave it unchanged. Without a progress condition or a positive separation from the tolerance threshold, the procedure has no general finite-stopping guarantee.

For the integer-CFLP model, binary openings are fixed first; the remaining demand equalities and facility capacity inequalities form a transportation/network-flow polyhedron with integral right-hand sides. An integral optimal extreme flow exists, so integer opening and unit shipping costs make every prefix *optimum* integral. Feasible fractional flows may have fractional objectives; the old shorthand "every feasible solution has integer cost" is false. Under exact arithmetic and valid $L \le J_B \le U$, integer endpoint projection is $\lceil L \rceil \le J_B \le \lfloor U \rfloor$. Under floating arithmetic, the direction of any outward correction to an asserted lower or upper bound must first be justified by the solver's numerical-error contract or independent verification. Arbitrarily applying `ceil` to a displayed lower bound is not a substitute for that justification.

The locked exact-oracle implementation checks `UB-LB`$< 1 - 0.05$ and uses `int(round(UB))`. Its full saved objective and gap are available for each label, but intermediate candidate bounds and near-integer distances were not persisted for all tested prefixes. Consequently this manuscript follows the numerically qualified Route B: it reports labels as solver-audited under that criterion and does not claim a per-prefix, outward-rounded exact-arithmetic certificate. The exact lattice corollary remains a mathematical statement conditional on truly valid intervals. This qualification does not modify the locked empirical coverage numbers.

E Complete Locked CFLP Confidence Grid

All four paths use the locked B.5 protocol and the same 20 held-out instances per scale. The table entries are mean seconds of instance-level medians of three calls, excluding separately executed MIP-start validation. A/B/C/D are defined in Section 8.6; A/D is the ratio of means, and the interval is a 95% paired bootstrap interval from resampling instances. "Initial full" and "recovery to full" are structural events, not objective certificates. All B.5 paths reached a solver-reported absolute-gap certificate; the LP yielded no early certificate.

Table 3: Complete pre-specified real-valued CFLP confidence grid. Candidate budgets target reference fidelity, not mathematical $Q_{\mathcal{C}}(\theta; 0)$.

N	α	B_1/N	Initial full	Recovery to full	A	B	C	D	A/D	95% CI
8	.20	.875	0%	100%	.0163	.0366	.0360	.0327	.499	[.414,.602]
8	.10	1.000	100%	0%	.0158	.0194	.0194	.0160	.987	[.948,1.031]
8	.05	1.000	100%	0%	.0158	.0194	.0194	.0160	.992	[.943,1.037]
20	.20	.750	0%	100%	.0517	.0981	.0937	.0879	.588	[.521,.669]
20	.10	.800	0%	100%	.0568	.1119	.1119	.1054	.539	[.461,.625]
20	.05	.950	0%	100%	.0528	.1147	.1076	.1029	.513	[.471,.560]
30	.20	.733	0%	100%	.0881	.1664	.1584	.1519	.580	[.525,.638]
30	.10	.767	0%	100%	.0878	.1678	.1611	.1507	.583	[.519,.653]
30	.05	.867	0%	100%	.0880	.1785	.1674	.1589	.554	[.503,.607]
50	.20	.740	0%	100%	.1359	.2711	.2525	.2378	.572	[.524,.630]
50	.10	.780	0%	100%	.1334	.2747	.2552	.2411	.553	[.502,.614]
50	.05	.900	0%	100%	.1319	.2856	.2619	.2472	.533	[.496,.583]

F Portfolio Formulation and Secondary Computational Details

At a decision epoch, let $x^-_{h,i}$ and $x_{h,i}$ be pre- and post-trade risky weights, $u_{h,i} = x_{h,i} - x^-_{h,i}$ signed trade, $q_{h,i} \geq |u_{h,i}|$ absolute-trade auxiliary, $y_{h,i}$ holding binary, $z_{h,i}$ charged-trade binary, and b^-_h, b_h cash weights. With horizon $h = 0, \ldots, H-1$, the audited MIQP minimizes discounted risk minus expected return and cash yield plus transaction cost:

$$
\begin{aligned}
\min \; & \sum_{h=0}^{H-1} \beta^h \left[\tfrac{\gamma}{2} x_h^\top \Sigma_h x_h - \mu_h^\top x_h - r_{f,h} b_h + \kappa_h \right] && (13)\\
\text{s.t. } & x^-_0 = \bar{w}_t, \quad b^-_0 = \bar{c}_t, \quad x_h = x^-_h + u_h, \\
& \mathbf{1}^\top x_h + b_h = \mathbf{1}^\top x^-_h + b^-_h - \kappa_h, \quad \kappa_h = \sum_i (a_{i,h} q_{h,i} + f_{i,h} z_{h,i}), \\
& q_{h,i} \geq u_{h,i}, \quad q_{h,i} \geq -u_{h,i}, \quad \ell_{i,h} y_{h,i} \leq x_{h,i} \leq U_{i,h} y_{h,i}, \quad \sum_i y_{h,i} \leq K_h, \\
& \delta_{i,h} z_{h,i} \leq q_{h,i} \leq M^q_{i,h} z_{h,i}, \quad \sum_i q_{h,i} \leq \tau^{\max}_h, \quad 0 \leq q_{h,i} \leq L_{i,h}, \\
& x^-_{h+1,i} = g_{i,h} x_{h,i}, \quad b^-_{h+1} = (1 + r_{f,h}) b_h \quad (h < H-1), \\
& b_h \geq b^{\min}_h, \quad x^-_{h,i} \geq 0, \quad y_{h,i}, z_{h,i} \in \{0,1\}.
\end{aligned}
$$

The first planned trade is executed at each weekly step and the model is resolved. Entity exclusion fixes all corresponding risky-asset block variables to their null values; global cash remains. For the controlled formulation, a physical candidate budget B yields $H(6B+3)$ variables, including $2BH$ binaries, and $H(8B+5)$ counted linear constraints under the implementation's formulation convention. These dimension identities do not imply a runtime complexity guarantee.

Perspective relaxation. For a positive-semidefinite risk matrix, a nonnegative diagonal D with $\Sigma - D \succeq 0$ permits the standard perspective treatment of diagonal terms in the continuous relaxation, while retaining the residual quadratic form. The strengthened relaxation is contained in the basic relaxation, hence for minimization $L_{\text{basic}} \leq L_{\text{perspective}} \leq J^*$. This is a standard deterministic lower-bound dominance statement; it does not guarantee improved total runtime. Archived portfolio runtime aggregates are not used to infer a speedup in this manuscript.

The archived chronological portfolio backtest is not used for an investment-efficacy claim, a common-state solver-time comparison, or a candidate-quality label. Its legacy aggregate columns remain in the repository for provenance but are outside the inferential evidence of this paper.

G Reproducibility and Artifact Provenance

The real-valued CFLP calibration labels, raw Phase B records, exact-oracle labels and summaries, and timing comparisons are under `artifacts/k20_replication/`. SHA-256 hashes of the Phase B and B.5 raw CSVs match the saved manifests. The read-only original LP-barrier calculation subtracts `lp_valid_bound` from `reference_solver_bound` in each of the 240 Phase B rows and compares it with `certificate_tolerance=0.001`. B.5 corroboration uses `full_lb-lp_lb` in the 1,440 rows with an LP field. This barrier calculation did not rerun any LP or alter a locked outcome. The separate T6.1 full-prefix output and manifest reside under `artifacts/t6_1_full_prefix_reanalysis/`; they record new solver runs on exactly the same 80 instances and are not pooled with original test statistics.

The final four-path timing uses Gurobi 13.0.3 on Windows 11 with an AMD Ryzen 7 170 with Radeon Graphics and approximately 16 GB RAM, one solver thread, the above absolute gap target, and instance-paired resampling, as recorded in the B.5 manifest. Phase B and the separate integer-oracle experiment have their own saved solver settings and should not be

pooled as a single timing protocol. The original Phase B and B.5 manifest files specify their scripts, environment, seeds, and artifact hashes. The new T6.1 manifests additionally hash the generator/solver code, 1,414 raw prefix rows and analysis outputs, while retaining each locked generation seed and reconstructed instance fingerprint. Historical intermediate-time bound traces remain unavailable; final prefix-bound identification is therefore not an early-time solver measurement.

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